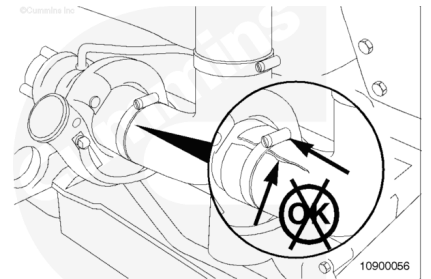


Trainings ▾

Initial Check

⚠ CAUTION ⚠

Engine intake air must be filtered to prevent dirt and debris from entering the engine. If intake air piping is damaged or loose, unfiltered air will enter the engine and cause premature wear.



Inspect the intake air piping for cracked hoses, damage, or loose clamps.

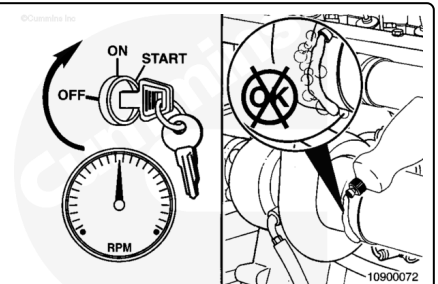
Replace damaged pipes, and tighten loose clamps, as necessary, to make sure the air intake system does **not** leak.

Torque Value: 8 n·m [71 in-lb]

Check for corrosion of the intake system piping under the clamps and hoses. Corrosion can allow corrosive products and dirt to enter the intake system. Disassemble and clean, as required.

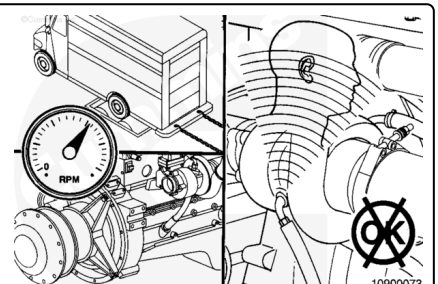
Operate the engine at high idle, and use a solution of soapy water to spot intake air leaks.

If an air leak exists, the soap bubbles will be drawn in with the air.



Operate the engine at full throttle and rated rpm with maximum load.

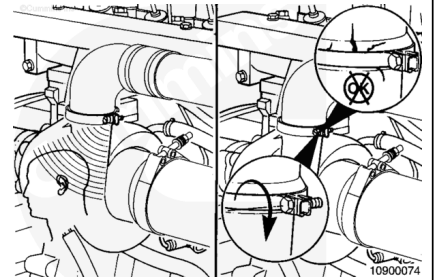
Listen for a high-pitched whistling noise from the turbocharger, nearby piping, and connections.



The noise can be caused by an air leak from the:

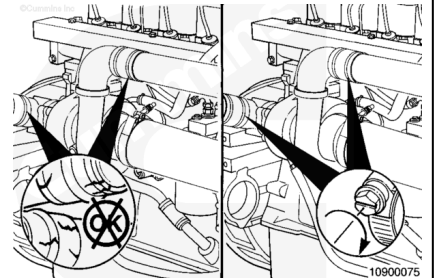
1. Turbocharger-to-discharge elbow connection.
 - Inspect for damage. Tighten loose clamps.

Torque Value: 8 n•m [71 in-lb]



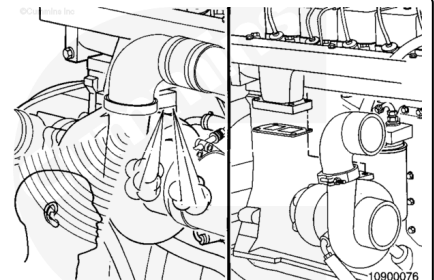
1. Any charge air cooler tubing or connecting hoses.

- Inspect the hose and tubing for damage.
- Tighten the hose clamps.
- Refer to the manufacturer's specifications for the correct torque value.



1. Turbocharger-to-exhaust-manifold mounting gasket.

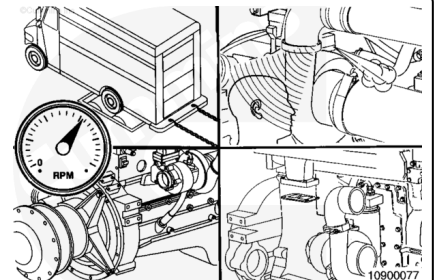
- Replace the gasket. Refer to Procedure 010-033 (</qs3/pubsys2/xml/en/procedures/40/40-010-033-tr.html>).



Operate the engine at full throttle and rated rpm with maximum load.

Listen again for leaks.

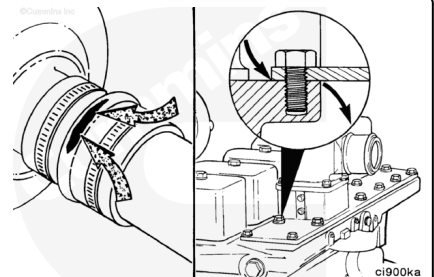
Replace the turbocharger if the air piping is **not** damaged and the noise can still be heard. Refer to Procedure 010-033 (</qs3/pubsys2/xml/en/procedures/40/40-010-033-tr.html>).



Damage from Nonfiltered Air

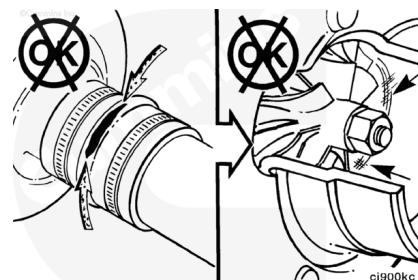
Loose connections or cracks in the suction side of the intake pipe and after the air filter can allow debris to be ingested by the engine, causing rapid wear in the cylinders.

Leaks at the intake manifold, unsealed bolt holes, or manifold cover gasket can also allow dust and dirt to be ingested into naturally aspirated engines.



Debris drawn into the air suction side can damage the compressor blades, causing an imbalance resulting in bearing failure.

To verify a bearing failure or damaged compressor, remove the intake and exhaust piping, and check for contact. The rotor assembly **must** rotate freely and should **not** be damaged. Measurement of axial and radial clearance is described in this section.



Turbocharger Engines - Air Leaks, Pressure Side

Excessive smoke and low power from a turbocharged engine can be caused by pressurized air leaking from loose connections or cracks in the crossover tube or intake manifold. This can also cause a noise problem.



In addition to the inspection for cracks and loose fittings, liquid soap can be applied to the charge air cooler, connections, and the manifold cover sealing surfaces to find the leaks. The leaks will create bubbles that are easier to detect. Measurement of manifold pressure is described in this section.

